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# Jan Exley

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## EDUCATION

2022 - Present **The University of Manchester**
**MMath&Phys Mathematics and Physics (Integrated Masters)**
1st yr result: 76.9%, 2nd yr result: 77.6%, 3rd yr result: 80.3%

- **1st Semester Master's Project:** Optimisation Algorithms for Computerised Tomography (inverse problems, Radon transforms, PDHG optimisation) **Supervisor: Prof. Sean Holman**
- **2nd Semester Master's Project:** Quantum Computing in Astronomy - Investigating superconducting qubit architectures, cryogenic systems & thermal modelling for quantum device characterisation. **Supervisor: Prof. Lucio Piccirillo**
- Completed advanced Quantum Mechanics modules including Quantum Information & Computation (85%), Advanced Quantum Mechanics (80%), with further study in Gauge Theory ongoing.
- Strong performance in modules such as Linear Algebra (85%), Electrodynamics (85%), Differential Geometry (90%), PDEs & Vector Calculus (90%), Complex Analysis (86%), Nuclear & Particle Physics (87%).
- Applied advanced Mathematical and Physical concepts in programming modules such as OOP in C++ (86%).

### PROJECTS AND EXPERIMENTAL WORK

- Publicly Showcased Group Project titled "Can we travel to Tokyo from Manchester in 40 Minutes?" - Numerical optimisation problem.
- Yr2 Lab Work – 72% - Calculated galaxy distances using data from Hubble Space Telescope. Optimised Triode Valve configuration by running a series of experiments. Implemented an RFID reader & tested other sensors using the microcontroller Arduino. All experiment findings were presented & defended to academic professionals.
- Professional Development presentation titled "To Build a Quantum Computer, why?".
- Yr1 Lab Work – 72% - Explored oscillating systems, the gravitational constant, projectile motion, fundamental data science. Included lab report interviews.
- Special Topics Group Work **-** Designed & presented an informational booklet on the Large Hadron Collider at CERN as part of a team to a High School readership.

2021 - 2022 **The University of Manchester**
**BSc Computer Science and Mathematics**

- Ended the degree after a year as it did not align with academic & research interests. Gained experience in collaborative software development, data science, introductory AI & statistical modelling.

2016 - 2021 **Bradfield College (House Prefect for Sixth Form)**

- **A-Levels:** Mathematics(A*), Computer Science(A*), Further Mathematics(A), Physics(A), Extended Project Qualification(B).

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- **GCSEs:** Maths (9), Physics (8), Chemistry (8), Latin (A*), English Language (7), plus additional subjects graded 9–6.

## ENGINEERING & TECHNICAL EXPERIENCE

2024 – **Industry Exposure & Mentoring** – Visited Waters Corporation research facility & met senior research scientist, Keith Richardson & had multiple follow-up mentoring meetings. Gained exposure to ion-trap based instrumentation, RF field control & precision charged-particle measurement. Developed valuable insight into the collaboration between industry & academia.

2022 - 2023 - **Hyperloop Manchester - Sub-Team Member** - Developed a working magnetic levitation system for Elon Musk's high-speed transportation system to be showcased at European Hyperloop Week.

2021 – **EPQ -** Produced essay on "What can we do to reduce our house's ecological footprint" which analysed different possible solutions, such as solar power & heat pumps to reduce the carbon footprint of the house.

## WORK EXPERIENCE

06/23 - Present - **MyTutor - Online Tutor -** Providing personalised tutoring lessons to students from KS3 to university level. Dedicated to encouraging & enlightening students about STEM.

06/22 - 09/24 - **Oakley Hall - Bartender & Waiter -** Delivered customers with a smooth & enjoyable experience, customising cocktails & offering efficient service. Trusted with closing procedures, including end-of-day cash balancing & managing stock. Leading other less experienced team members.

## LEADERSHIP ROLES

07/24 - Present - **UOM Futsal Society/Club - Social Media Sec -** Handling social media presence of the club with a view to boosting participation & to attract sponsors.

04/23 - 07/24 - **PASS Leader -** Provided support for incoming 1st year students, through weekly study sessions to aid integration into university life.

07/23 - 07/24 - **UOM Futsal Society/Club - Treasurer -** Managed financial affairs of the club. Creating membership plans, forecasting expected membership intake, determining additional costs, devising a debt recovery plan, organising fundraisers, writing employee contracts, persuading potential sponsors.

## ADDITIONAL SKILLS

**Programming:** Python, C++, MATLAB, LaTeX, Git, Linux; experience with Java, SQL, web technologies.

**Languages:** English (Native), Polish (Native), French & Spanish (Conversational).

## INTERESTS AND OTHER ACTIVITIES

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**Societies:** Active member of the recreational UAV, Robotics, Physoc & Mathsoc societies. From making drones to participating in science quizzes.

**Music & Drama:** Saxophone (grade 7), Sing (grade 7), LAMDA Solo Acting (grade 7). Previous member of Bradfield College's Big Band & Saxophone Ensemble, playing in charity concerts. Participated in school drama productions, including performing a Greek play in Greek!

**Other Activities:** Alongside representing Manchester University in futsal I play dodge ball, village cricket & squash. I am a massive Star Wars & Lego fan. Completed various charitable fundraising activities, including pushing my sister in a wheelbarrow for 5km! Other interests include dog walking, mountain biking, gym, reading, music & theatre.